SUMMIT BIKRAM NAYAK

Data Scientist | Machine Learning Engineer | Quantitative & Computational Modeler

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PROFESSIONAL SUMMARY

Quantitative Data Scientist and Machine Learning Researcher with an Integrated M.Sc. in Physics and a Minor in Computer Science from NISER (HBNI). Proven expertise in architecting end-to-end Python data pipelines, Bayesian probabilistic modeling (MCMC/HMC), differentiable programming (JAX, NumPyro), and High-Performance Computing (64-core parallelization). Skilled in translating noisy, high-dimensional datasets (hyperspectral imaging, biomedical vision, and large-scale numerical simulations) into scalable predictive models.

TECHNICAL SKILLS

Languages & Core:	Python (Advanced), C++, SQL, Bash/Shell Scripting, Fortran, IDL, MATLAB, LaTeX
ML & Data Science:	JAX, NumPyro, Scikit-Learn, NumPy, SciPy, Pandas, Numba, OpenCV, Feature Engineering
Statistical Modeling:	Bayesian Inference, Hamiltonian Monte Carlo (NUTS/MCMC), Time-Series & Spectral Analysis, Inverse Problems, Non-Linear Optimization, High-Dimensional Similarity Search (LSH)
Data Engineering & HPC:	Multi-Core Parallel Computing (64-Core HPC Clusters), Docker, Git, GitHub, GitLab, Linux/Ubuntu
Domain Expertise:	Hyperspectral & Geospatial Analytics, Remote Sensing, Computer Vision, Numerical Simulation

RESEARCH & DATA SCIENCE EXPERIENCE

Post-Graduate Researcher (Data Science & Probabilistic Modeling) **May 2025 – Aug 2025**
National Institute of Science Education and Research (NISER) *Bhubaneswar, India*

- Architected fully differentiable forward models and Bayesian inference pipelines using **JAX** and **NumPyro** to extract latent physical parameters from noisy, high-dimensional hyperspectral datasets.
- Engineered parallelized data processing workflows across **64-core High-Performance Computing (HPC)** clusters, drastically reducing runtime for complex parameter retrieval and resolving severe multi-variable degeneracies.
- Automated end-to-end spectral ingestion and predictive modeling pipelines, generalizing legacy codebase architectures for scalable cross-domain deployment.

Science Team Contributor (Computational Atmospheric Modeling) **Aug 2025 – Present**
L.I.F.E. Collaboration (Large Interferometer For Exoplanets) *Remote*

- Collaborating with an international data science and astrophysics consortium to develop synthetic spectral datasets and statistical retrieval benchmarks for planetary characterization.

Graduate Machine Learning Researcher (Master's Thesis) **Jul 2024 – May 2025**
National Institute of Science Education and Research (NISER) *Bhubaneswar, India*

- Built an automated Python and Fortran data pipeline to process large-scale satellite hyperspectral data cubes (VIRTIS-M-IR), implementing multi-stream radiative transfer and scattering algorithms.
- Replaced fragile deterministic optimization routines with **Hamiltonian Monte Carlo (NUTS)** sampling, delivering rigorous Bayesian uncertainty quantification and posterior distributions.
- Conducted numerical truncation diagnostics and sensitivity analyses across high-dimensional lookup tables (HITEMP2024), preparing a first-author technical manuscript on scalable retrieval architectures.

Computational Modeling Intern (Python & Density Functional Theory) **May 2023 – Jul 2023**
School of Physical Sciences, NISER (Advisor: Dr. Prasanjit Samal) *Bhubaneswar, India*

- Developed modular Python simulation scripts to automate many-body quantum mechanical calculations and analyze multi-variable convergence behavior.

Numerical Simulation & C++ Intern **May 2022 – Jul 2022**
Indian Institute of Technology (IIT) Mandi (Advisor: Dr. Prasanth P. Jose) *Mandi, India*

- Designed high-efficiency **C++** numerical solvers to simulate 3D dynamical systems and performed statistical data analysis on large simulation outputs.

KEY MACHINE LEARNING & DATA CHALLENGES

Bio-Inspired Locality-Sensitive Hashing (LSH) for High-Dimensional Similarity Search *Jan 2025 – May 2025*

- Implemented and optimized a modified Fly Olfactory Locality-Sensitive Hashing (LSH) algorithm in Python for fast approximate nearest-neighbor (ANN) similarity search across high-dimensional feature spaces.
- Benchmarked retrieval precision and computational efficiency against standard hashing baselines under Dr. Subhankar Mishra (School of Computer Sciences, NISER).

Microlensing Data Challenge (MLDC) — HPC Time-Series Modeling *May 2026*

- Competed as a solo data scientist (`mldc-nayak-niser-solo`), deploying parallelized parameter estimation algorithms on HPC clusters to model complex gravitational microlensing time-series light curves.

Automated Computer Vision Pipeline for Biomedical Lipid Quantification *Jul 2023 – Dec 2023*

- Developed an automated image processing and segmentation pipeline in Python and ImageJ to quantify cellular lipid content from stained microscopy images, performing statistical pre/post-treatment comparisons.

EDUCATION & CERTIFICATIONS

National Institute of Science Education and Research (NISER / HBNI) **Aug 2020 – May 2025**
Integrated M.Sc. in Physics (Major) with Minor in Computer Science | CGPA: 6.80/10.0 *Bhubaneswar, India*

- **Relevant Coursework:** Machine Learning, Advanced Machine Learning, Design & Analysis of Algorithms, Programming & Data Structures, Parameterized Algorithms, Discrete Structures, Theory of Computation, Linear Algebra, Statistical Mechanics, Computational Physics.

Certifications & Specialized Training:

- **AI/ML for Geodata Analytics** — Indian Institute of Remote Sensing, ISRO (Aug 2026)
- **Astrobiology & Planetary Data Analysis** — RED Astrobiology School, France (May 2026)

HONOURS & TECHNICAL PRESENTATIONS

- **DISHA Fellowship (2020 – 2025):** Awarded 5-year merit scholarship by the Department of Atomic Energy (DAE), Government of India.
- **Invited Participant, ISRO Venus Orbiter Mission (VOM) Meet (Nov 2025):** Selected to attend technical calibration and payload data processing sessions at ISRO HQ, Bengaluru.
- **Research Presenter, S2 STEP Conference (Feb 2025):** Presented computational spectral retrieval pipeline at IIT Roorkee.